

Assignment of Chapter Three (B)

A. FILL-IN-BLANK

1. When designing multiple output combinational logic circuits, we need to fully consider the _____, so that the circuits can be further simplified.
2. The relationships of a combinational logic circuit can be represented by _____, and _____, such two methods.
3. The methods to determining whether a circuit can have hazards include _____, and _____.
4. When designing a combinational logic circuit with “don’t care” terms, we must make the full use of the arbitrariness of the “don’t care” terms. This statement is _____ (True/False).
5. The race phenomenon is caused by incorrect circuit designs. This statement is _____ (True/False).
6. In combinatorial logic circuits, a hazard caused by race is a transient error that does not cause the circuit’s a steady state result. This statement is _____ (True/False).

B. CIRCUIT ANALYSIS

Whether the combinational circuits presented by the following three algebra expressions can produce races, and which conditions can lead the races to hazards?

1. $F_1 = AB + A\bar{C} + \bar{C}D$

2. $F_2 = AB + \bar{A}CD + BC$

3. $F_3 = (A + \bar{B})(\bar{A} + \bar{C})$

C. CIRCUIT DESIGN

1. Use the NAND gates to implement a combinational logic circuit, whose input is a one-bit 2421 code number. The circuit is to determine whether the input is a prime number; if so, $F=1$, if not $F=0$.
2. Use the NAND gates to design a rounding function circuit, whose input is a one-bit 8421 code number. If the number is larger than 5, $F=1$, if not $F=0$.

3. Design a “FULL ADDER/ SUBTRACER”, it has a input M, which control the circuit’s addition and subtraction processes. If M=0, the circuit is a FULL ADDER; if M=1, the circuit is a FULL SUBSTACER.

(Tips: the circuit has 4 inputs, M, A, B, C; A denotes the summand/ minuend; B denotes addend/minuend; C denotes for the carry/borrow from the lower bit; F denotes the sum/difference; G denotes the carry/borrow to the high bit)